CHIH-HAO (ANDY) TSAI

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Github Portfolio: https://github.com/andytsai104/andytsai-robotics-portfolio

Education

M.S. in Robotics and Autonomous Systems — Aug. 2024 – May 2026
Arizona State University, Tempe, Arizona, United States — GPA: 3.5/4.0

Study Abroad Program in Electrical Engineering — Mar. 2023 – Aug. 2023
Aachen University of Applied Sciences, Aachen, Germany — GPA: 3.3/4.0

B.S. in Mechanical Engineering — Sep. 2019 – Jun. 2023
National Taipei University of Technology, Taipei, Taiwan — GPA: 3.03/4.0

Technical Skills

- **Programming: Python** (Advanced – PyTorch, TensorFlow, OpenCV), **C**, **C++**, **MATLAB** (Intermediate – kinematics, control), **Bash** (Intermediate – Linux automation/HPC automation)
- **Robotics & Systems: ROS 2**, **Nav2**, **URDF/xacro**, **Multi-Robot Systems**, **Motion Planning**, **Path Planning**, **PID/PI Control**, **Odometry/IMU Integration**
- **Tools & Simulation: Linux**, **Git**, **RViz**, **Gazebo** (Fortress/Harmonic), **CARLA**, **Simulink**, **SolidWorks**
- **Machine Learning: Deep Learning**, **Reinforcement Learning**, **Computer Vision**, **Vision-Language-Action Models**, **Behavior Cloning**

Professional Experience

BELIV Lab, Arizona State University — Jun. 2025 – Present
Research Assistant — Mesa, Arizona

- Built a **BC+TD3** pedestrian policy-learning framework in **CARLA** for safety-critical AV scenario generation.
- Trained **Behavior Cloning** models on **ASU SOL** using **Bash** scripts for **HPC job automation** and experiment tracking.
- Designed semantic **BEV-based** observations with kinematic and goal features for continuous pedestrian control.
- Improved safety-critical interactions, increasing collision rate from **0%** to **45%** and reducing minimum vehicle distance.

Academic Projects

OmniVLA Semantic Warehouse Navigation (*Team Leader*) — Jan. 2026 – Apr. 2026

- Built semantic warehouse navigation with **OmniVLA-edge**, **ROS 2 Jazzy**, **Gazebo Harmonic**, and **Nav2**.
- Developed a semantic goal library with language aliases, goal images, and warehouse navigation poses.
- Built a **ROS 2** data collection and JSONL export pipeline for **VLA** fine-tuning.
- Fine-tuned **OmniVLA-edge** for semantic goal prediction, reaching **97.57% validation accuracy** and about **80% evaluation accuracy**.

Consensus-Based Multi-Robot Warehouse Task Allocation *(Team Leader)* — Aug. 2025 – Dec. 2025

- Built **Max-Consensus** task allocation with distance-based bidding and one-task-per-robot assignment.
- Implemented a **ROS 2 Humble + Gazebo Fortress** warehouse simulation with **2D LiDAR** robots.
- Connected allocation results to **Nav2** goal dispatch and task-status monitoring.
- Verified **finite-time convergence** and conflict-free assignments under capacity constraints.

Pololu 3pi+ 2040 Embedded Robotics *(Individual Project)* — Aug. 2025 – Dec. 2025

- Built autonomous behaviors using IMU, encoders, gyroscope, IR line sensors, bump sensors, and LCD telemetry.
- Implemented line following with IR calibration, lateral-error estimation, P control, and differential steering.
- Developed ramp edge-detection and recovery logic with backup/turn maneuvers for left, center, and right edges.

Vision-Based Maze Solving & Path Planning with MyCobot Pro 600 (*Team Leader*) — Mar. 2025 – Apr. 2025

- Developed a **ROS 2-based pipeline** to control a 6-DOF robotic arm using camera-captured paths.
- Built a digital twin (URDF) with **SOLIDWORKS** for simulation in **RViz** and **Gazebo**.
- Applied **OpenCV** in Python to process maze images, including path extraction and skeletonization.
- Executed joint trajectories on both simulation and physical robot via **TCP/IP**, optimizing motion smoothness.

Robot Forward/Kinematics (ROS2 & Gazebo & MATLAB) (*Team Leader*) — Feb. 2025 – Mar. 2025

- Built a simulation model in **ROS2, Gazebo and Solidworks** for the Dobot Magician Lite robotic arm.
- Simulated a SCARA robot and performed motion control in **Simulink**.
- Validated forward and inverse kinematics using MATLAB and Python scripts.

Autonomous Mobile Vehicle and Robotic Arm (*Team member*) — Feb. 2022 – Nov. 2022

- Built an autonomous mobile vehicle with a robotic arm for object relocation in a 4-person team.
- Led webcam-based object detection using **TensorFlow + OpenCV** (**Python**).
- Designed the vehicle/arm in **SolidWorks** for 3D printing and integrated motor/arm control on **Arduino (C)**.